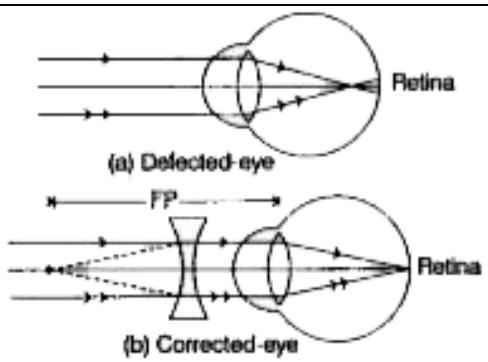
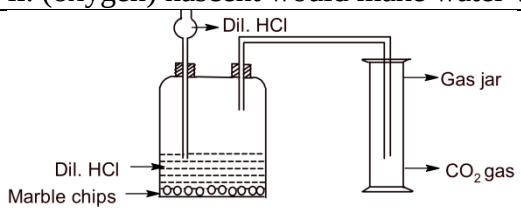
SEE 2081 (2025)

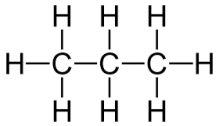
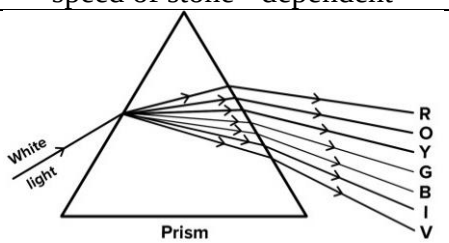
अनिवार्य विज्ञान तथा प्रविधि

उत्तरकुन्जिका

o; s'lGhsfdf ePsf] afx]s cGo ;xL j}slNks tl/sfaf6 ;d:of ;dfwfg u/]df klg c+s k|bfg ug'{kg]{ 5 . pQ/s'l~hsfdf k|To]s r/0fsf] k|fKtf+Í ! k|bfg ug'{kg]{5 tyf PsfO 5'6 ePdf)=% cª\s dfq 36fpg'kg]{ 5 .

k|= g+=	pQ/	cª\s
	Group A	
1) a.	(iv) 46	1
1) b.	(iii) $P=\frac{1}{f}$	1
1) c.	(iv) tricuspid valve	1
1) d.	(iv) Holy basil	1
1) e.	(ii) ethylene glycol	1
1) f.	(i) increases	1
1) g.	(ii) electric motor changes electrical energy into mechanical energy	1
1) h.	(iv) nature of matter	1
1) i.	(iii) kgm^2s^{-2}	1
1) j.	(i) 875 N	1
	Group B	
2) a.	Aorta	1
2) b.	Ammonium hydroxide (NH_4OH)	1
2) c.	because it is heavier than air	1
2) d.	Think before posting the material in social site Not to hurt anyone's personal life .or any other correct answer ..	1
2) e.	It decrease rate of chemical reaction	1
2) f.	non- biodegradable	1
2) g.	ratio is 1:1	1
2) h.		1
2) i.	Refractive index (μ)$=\frac{speed\ of\ light\ in\ air}{speed\ of\ light\ in\ medium}$ $=\frac{3\times10^8}{2.25\times10^5}$ =1333.33	1
	Group C	
3.	• Mouth parts are lapping and chewing type • It is relatively smaller than other bee.	1 1

4.	• global warming • off seasonal rainfall or other correct answer	1 1
5.	<table><tr><th>Independent</th><th>dependent</th></tr><tr><td>(i) It is manipulated by researcher (ii) it is cause</td><td>(i) It is not manipulated by reseracher ii) it is effect</td></tr></table> Any two correct difference will get full marks	1 1
6	sparrow : class: Aves fern: division pteridophyta (Tracheophyta)	2
7	• experiment cannot be controlled • it takes long time • it may not have number of contrasting characters as pea. or any other correct answer gets 1 mark of each	1 1
8.	• 16 N indicates upthrust of the liquid to the object or wt. of liquid displaced. • If the wt of the object is 16 N, it would float on the water	1 1
9.	<table><tr><th>Flat universe</th><th>Closed Universe</th></tr><tr><td>• the value of average density is equal to the critical density</td><td>• the value of average density is greater than the critical density</td></tr><tr><td>• the size of universe remain constant</td><td>• the size of this type of universe becomes smaller and collapse</td></tr><tr><td colspan="2">or any other correct answer gets 1 mark of each</td></tr></table>	1 1
10.	i. Combination reaction is Reaction A ii. (oxygen) nascent would make water with hydrogen	1 1
11.	 Fig: Laboratory preparation of CO_2 gas • calcium carbonate	1 1
12.	• to increase the productivity • to protect the plants from different kinds of diseases. any other correct answer gets 1 mark of each	1 1
13.	• Whale- most developeed • sea-horse - least developed	1 1
14.	700 g= 0.7 Kg change in temperature= 35^0C Q= msdt Q = 0.7 x 4200x 35 = 1,02,900 J or 102.9 KJ	1 1
15.	i. In part A ii. it acts as electrolyte	1 1

16.	 Propene	1 1
	Group D	
17.	**<u>S-Block</u>** • They have one or two electrons on their valence shell • the valence electrons end at s-subshell • soft metals • highly metallic or other correct answer **<u>P Shell block element</u>** • They have 1-6 number of electrons in the valence shell • electronic configuration ended at p-subshell • non metallic • most reactive non metal any two correct answers gets 1 mark of each.	1 1 1 1
18.	(i) independent - sunlight dependent - growth (ii) stretching - independent speed of stone - dependent	1 1 1 1
19.	 Dispersion of white light Reasons: (i) velocity of each colours of light is different (ii) each colour of light have different wave length or other correct answer	1 1 1 1
20.	(i) Y - platelets (ii) Leukepenia (iii) The blood does not clot during cuts and injuries (iv) RBC (Red blood cells)	1 1 1 1
21	i) Here, $V_p = 240$ v $V_s = 480$ v $N_p = 300$ $N_s = ?$ $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $Ns = \frac{N_P \times V_s}{V_p}$ $= \frac{300 \times 480}{240}$ $= 600$ required turns = 600 turns second ii) To reduce eddy current loss. iii) No, use of transformer is to change the output voltage but in DC there is no change in polarity.	1 1 1 1

22.	**advantages** • to produce advance and healthy hybrid • it is possible in the absence of male organism **precaution** • The male from which sperm is collected should be healthy • during storing of sperm we should be aware or other correct answer	1+1 1 1
23.	(i) $m_1 = 6.4 \times 10^{24}$ kg . m2 = 400 kg d = 6400 km = 6.4×10^6 m F = G m1xm2/d^2 $= \frac{6.67 \times 10^{-11} \times 400 \times 6.4 \times 10^{24}}{\left[(6.4)^6\right]^2}$ =3913.93 N (ii) g = GM/$(R+h)^2$ 6 = $6.67 \times 10^{-11} \times 6 \times 10^{24}$/ $(6.4 \times 10^5+h)]^2$ h= 1.7 $\times 10^6$ m or 1700 km.	1 1 1 1

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